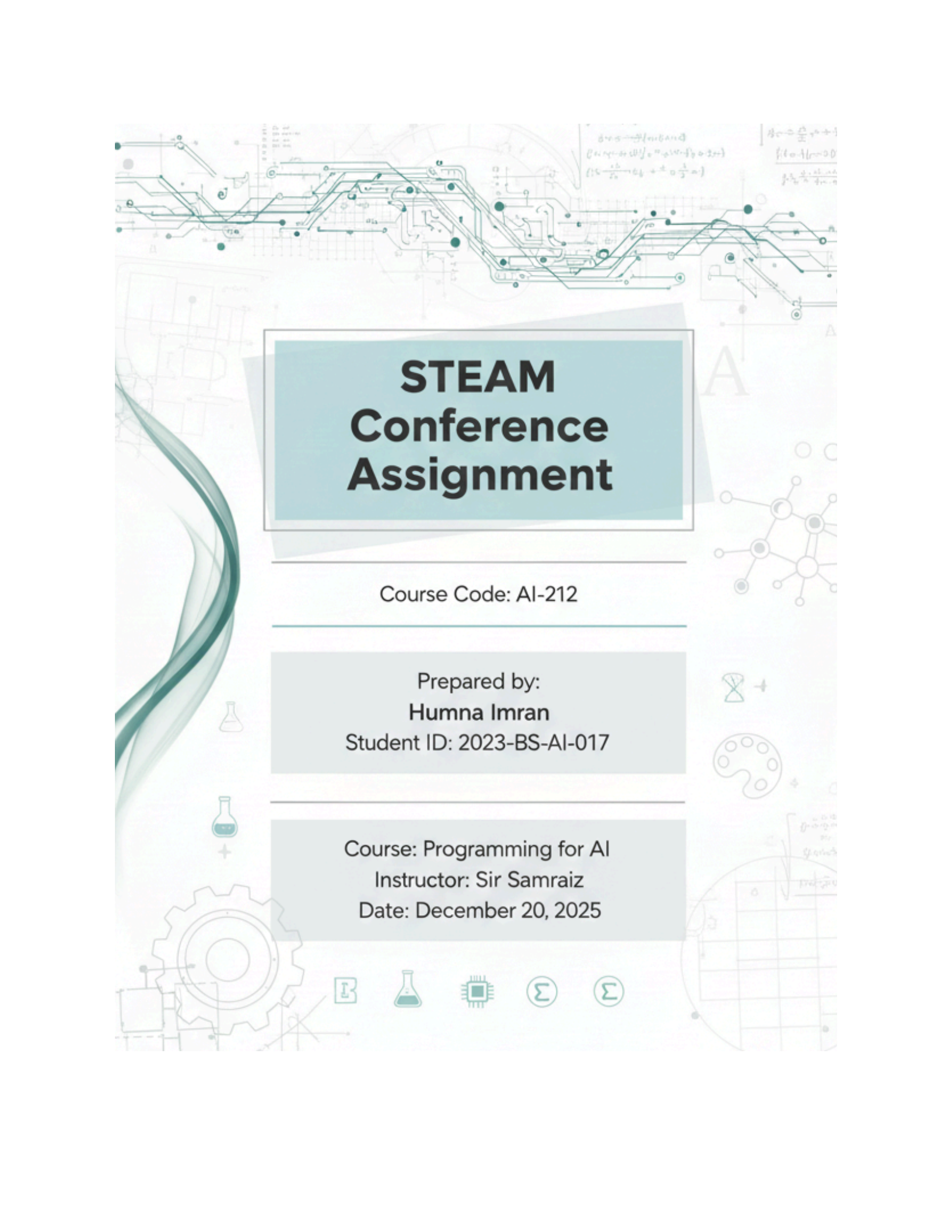


The background of the cover is a light green and white collage. It features a complex circuit board pattern at the top, a large gear at the bottom left, and various mathematical formulas and icons (like a flask, a palette, and a DNA helix) scattered throughout. A large, stylized letter 'A' is visible on the right side. The title 'STEAM Conference Assignment' is centered in a teal box.

STEAM Conference Assignment

Course Code: AI-212

Prepared by:
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Course: Programming for AI
Instructor: Sir Samraiz
Date: December 20, 2025

STEAM Conference Reflection Report

A. Introduction

This report reflects on my attendance at the 3rd International Conference on Advanced STEAM Education (ICASE-2025), hosted by The University of Faisalabad. The event served as a global gathering for scholars and industry leaders to discuss the intersection of Science, Technology, Engineering, Arts, and Mathematics. I attended all three days of the conference, which focused on the central theme of "*Connecting Technology, Arts and Social Science for Shaping the Future*." In this report, I will specifically discuss the sessions regarding Agentic AI led by Prof. Dr. Nadeem Javed, the Data Science workshops by Prof. Dr. Kamran Munir, and the collaborative scientific sessions on Federated Learning and healthcare AI. These sessions provided a deep dive into how interdisciplinary innovation is currently solving real-world problems.

B. Session Summaries & Analysis

Session 1: Autonomous Agents: Is Agentic AI the Next Evolution or an Illusion?

- **Speaker:** Prof. Dr. Nadeem Javed (Tamsuk, Taiwan).
- **Key Takeaways:**
 - The session explored the "*Eron Mark IQ*" metric, illustrating the leap from basic conversational AI (IQ 35) to the theoretical state of "*Complete*" AI (IQ 150,000).
 - A distinction was made between "*Automatic*" systems and truly "*Autonomous*" agents that possess goal-directed agency.
 - The speaker emphasized that while AI cognitive complexity is growing non-linearly, machines lack the emotional capacity for true human-like self-governance.
- **Critical Analysis:**

I found the differentiation between "*Intelligence*" and "*Wisdom*" particularly impactful. In our current studies, we often focus on the raw processing power of technology, but Prof. Javed's argument that machines are precluded from "*domination*" due to a lack of feelings offers a reassuring ethical boundary. I agree with the speaker's conclusion that Agentic AI is an evolution of intelligence, yet it remains a tool because it lacks the intrinsic emotional drive that characterizes human decision-making.

Session 2: Data Science Without Borders

- **Speaker:** Prof. Dr. Kamran Munir (UWE Bristol, UK).
- **Key Takeaways:**
 - Data science applications were showcased in industrial predictive maintenance, specifically regarding aircraft and rail asset monitoring.
 - The presentation highlighted sustainable agriculture through "*Agro Support*

Analytics" used to train farmers in Nigeria.

- The "*Colors of the Wind*" project demonstrated the integration of art and data science by using data to monitor environmental conditions.

- **Critical Analysis:**

The most innovative aspect of this session was the "*Colors of the Wind*" project. It perfectly encapsulated the "A" in STEAM by turning environmental data, which can often be dry or difficult to parse, into a medium for artistic and social awareness. It changed my perspective on data science from being a purely "*hard science*" to a tool for creative communication.

Session 3: AI, Data Science & Federated Learning Research Showcase

- **Speaker:** Chaired by Dr. Gufran Khan and Dr. Abdul Rauf.

- **Key Takeaways:**

- Presentations covered the use of deep learning for early lung disease detection and multimodal Alzheimer's prediction.
- The session explored "*Private local-language LLMs*," focusing on making AI accessible and secure for specific regional languages.
- Discussions included "*Explainable Legal NLP*," which aims to make AI legal decisions transparent and understandable.

- **Critical Analysis:**

The focus on "Explainable" AI in the legal sector was the most controversial yet necessary topic. As we move toward more automated systems, the "black box" nature of AI becomes a liability. I strongly agree with the researchers that for STEAM to be successful, it must be transparent; if we cannot explain why an AI reached a medical or legal conclusion, we cannot fully trust the technology.

C. Conclusion

The ICASE-2025 conference was a transformative experience that highlighted the necessity of international cooperation in education and technology. It moved beyond theoretical discussion into practical, life-saving applications in healthcare and environmental sustainability. My single most important takeaway from the event is that the future of technology is not just about computational power, but about "Human-Centric" design, ensuring that as our machines become smarter, they remain ethically grounded and accessible across cultural and linguistic borders.

D. Appendix: Proof of Attendance

